

A Housing and Additional Children: Evidence from Sibling Composition in the ACS

This appendix asks how the housing a young mother occupies changes when she has an additional child, using two features of her children that are plausibly unrelated to her housing preferences except through family size. The first is whether her two oldest children are recorded at the same age, which in most cases reflects a twin birth. The second is whether her two oldest children have the same sex, which Angrist and Evans (1998) show raises the probability of a third birth. Both sources of variation have a long history in the study of fertility and labor supply. Housing is a different outcome, and for the sibling-sex design an awkward one: children of the same sex can share a bedroom, so the composition of a family can change the housing it needs without changing its size.

The two designs study different families at different margins, and they give answers of opposite sign. Mothers whose two oldest children are the same age live in about a third of a room more than comparable mothers, and are about three percentage points more likely to own their home. Mothers whose two oldest children have the same sex live in slightly *fewer* rooms, a difference almost entirely accounted for by bedrooms. Taken at face value, the corresponding instrumental-variable estimate says that a third child reduces a family’s dwelling by more than a full room. We read this estimate as evidence that the exclusion restriction fails for housing outcomes, not as an effect of children, and we explain below why a small direct effect of sibling sex on housing is enough to produce it.

A.1 Data and the mother sample

The data are the one-year samples of the American Community Survey (ACS) for 2005–2019, covering the fifty states and the District of Columbia, as distributed by IPUMS USA. Each survey year is a separate cross-section of households. A person is identified by the survey year, the sample, the household serial number, and the person’s number within the household; a household is identified by the first three of these. No person is followed across years, so every quantity below is measured once, at the interview.

The ACS does not ask women how many children they have borne; it asks only whether a woman gave birth in the past twelve months, and it records no birth histories. We therefore reconstruct each woman’s children from the household roster. IPUMS assigns to each person the person number of a probable mother living in the same household, using relationship, age, and marital information (IPUMS USA, 2026). These links identify social as well as biological relationships: a stepmother or adoptive mother is linked in the same way as a birth mother. We keep every link in which the child’s age is nonnegative and below the mother’s age. The household relationship variable, which records each person’s relationship to the householder rather than to the mother, cannot separate biological from other links and is not used to filter them. Among all valid links in the 2005–2019 records, before any restriction on ages, 91 percent of linked children are recorded as the householder’s child and 4 percent as the householder’s grandchild, the latter typically a young mother living in her own parents’ home. Only 1,692 links fail because the recorded child is not younger than the recorded mother.

For each woman we count her linked coresident children, n_i , and order them by age, $a_{i1} \geq a_{i2} \geq \dots$, where ages are whole years at the interview. The count n_i is the number of her children living with her, not the number she has borne. Children who have left home, died, or live with another parent are missing, and a woman can be linked to a child she did not bear. Following Angrist and Evans (1998, pp. 452–453), we limit these errors by studying young mothers: the base sample consists of women aged 21 to 35 at the interview whose oldest linked child, if any, is younger

than 18. Angrist and Evans additionally drop mothers whose count of coresident children differs from their reported children ever born and mothers whose children’s ages or sexes were imputed by the Census Bureau. The ACS does not collect children ever born, and our extract does not carry imputation flags, so neither restriction is applied here.

Table 1 reports the resulting samples. The 59,046,776 person records in the source files fall to 46,373,936 once the sample is limited to the 2005–2019 one-year files. The base sample contains 4,024,238 women living in 3,783,862 households. It includes women with no linked child, who enter neither design, and it counts women rather than mothers.

Table 1: Sample construction

	Count
<i>Source records</i>	
ACS person records, fifty states and the District of Columbia	59,046,776
in the 2005–2019 one-year samples	46,373,936
Women aged 21–35 with positive person weight and no linked child aged 18 or older	4,024,238
households containing at least one such woman	3,783,862
<i>Same-age design</i>	
Mothers whose oldest linked child is aged 0–5	857,607
with two or more linked children	325,171
whose two oldest linked children are the same age	16,496
whose three oldest linked children are the same age	388
<i>Same-sex design</i>	
Women with two or more linked children	1,132,295
whose two oldest linked children are the same age	32,186
Mothers meeting all criteria (text)	656,986
two oldest children of the same sex	329,974
two boys	173,813
two girls	156,161
three or more linked children	148,775

Notes: Counts of person records, unweighted. Nested entries are subsets of the entry above them. In the same-sex design, the count of women whose two oldest children share an age is taken over all women with two or more linked children, whatever the age of the second child; the criteria for the estimation sample are listed in Section A.3.

A.2 Housing outcomes

We measure three features of the dwelling, all reported once per household and assigned to every mother in it. The universe of each is housing units outside group quarters.

Rooms counts whole rooms used for living purposes, including kitchens and excluding bath-rooms, porches, balconies, foyers, halls, and half-rooms. IPUMS codes valid counts from 1 to 27 and a top category of 30. In the 2009 file, 107 person records carry the value 28, which the code-book does not define; they are treated as missing. We cap the count at nine rooms because the Census Bureau imposed a national top code of nine rooms through the 2007 survey (IPUMS USA, 2026). Without the cap, a family’s measured size would depend on the survey year in which it was interviewed.

Bedrooms counts rooms that would be listed as bedrooms if the unit were for sale or rent, whatever their current use. The IPUMS code minus one is the number of bedrooms; valid codes run from 1 (no bedroom) to 22. We cap the count at five, again because a national top code of five bedrooms applied through 2007. Bedrooms are a component of rooms, and we report them to see

which part of the dwelling moves.

Ownership equals one if the unit is owned or being bought, including units held with a mortgage, and zero if it is rented, including units occupied without payment of cash rent. Other codes are missing.

Every mother in both estimation samples has a valid tenure code and a valid bedroom count. One mother in the same-sex sample lacks a valid room count, which is why the room samples in Table 2 are one observation smaller.

A.3 The two instruments

Same-age oldest pair. This design studies the move from one to two children among mothers whose first child is young. The sample consists of mothers with at least one linked child whose oldest linked child is aged 0 to 5. The instrument and the treatment are

$$Z_i^A = \mathbf{1}\{n_i \geq 2, a_{i1} = a_{i2}\}, \quad D_i^A = \mathbf{1}\{n_i \geq 2\}, \quad (1)$$

so the instrument indicates that the two oldest children are recorded at the same age and the treatment indicates a second coresident child. Every mother with $Z_i^A = 1$ has $D_i^A = 1$ by construction. The event age is the oldest child's age, $e_i = a_{i1}$, which measures the time since the first birth in whole years.

The instrument is a proxy for a twin first birth, not a twin indicator. Ages are recorded in whole years and our extract carries no quarter of birth, so two children born within a year of each other can share an age at the interview. A step-sibling or adopted child of the same age produces the same record. Angrist and Evans (1998, p. 454) measure the resulting error in the 1980 Census: defining twins by age in years alone yields a second-birth twin rate 35 percent higher than defining them by age and quarter of birth, and for that reason they restrict their twins analysis to 1980. In our sample 1.9 percent of eligible mothers have a same-age pair. Genuine twin births are themselves not random: Angrist et al. (2010, p. 775) note that twinning varies with maternal age and race and that twin births change child spacing and child health.

Same-sex oldest pair. This design studies the move from two to three children. The sample consists of mothers with at least two linked children whose two oldest children have different ages, whose second-oldest child is aged 0 to 5, and whose two oldest children both have a recorded sex. The instrument and treatment are

$$Z_i^B = \mathbf{1}\{s_{i1} = s_{i2}\}, \quad D_i^B = \mathbf{1}\{n_i \geq 3\}, \quad (2)$$

where s_{ik} is the sex of the k th oldest linked child. The event age is the second child's age, $e_i = a_{i2}$, so the clock starts at the second birth, when the instrument is realized. The requirement that the two oldest children differ in age removes the mothers who drive the first design and makes the order of the first two children well defined; the order of the second and third children is not checked, so when they share an age the child labeled second is arbitrary. The sample is not restricted to mothers with exactly two children, and it does not condition on the treatment. Every mother who satisfies the age and ordering criteria has a recorded sex for both children.

The instrument predicts a third coresident child, not a second, and its first stage is positive: parents of two boys or two girls are more likely to have another child. At event age 0 the second child was born within the past year, but a third child may already be present, so no event age in this design precedes treatment.

A.4 Estimating equations and inference

For each design, outcome Y_i , and window of event ages, we estimate a reduced form and a first stage on the same mothers:

$$Y_i = \delta Z_i + W_i' \gamma + u_i, \quad D_i = \pi Z_i + W_i' \kappa + v_i. \quad (3)$$

The reduced-form coefficient δ is the difference in housing between mothers with and without the sibling feature, and π is the corresponding difference in the probability of the additional child. The controls W_i are complete sets of indicators for the mother’s age at the event (her age at the interview minus e_i), her race, the survey year, and the event age. We do not control for the mother’s current marital status, education, or state of residence, all of which can respond to fertility; Angrist and Evans (1998, p. 465, footnote 12) exclude education and husband’s earnings for the same reason. Regressions are weighted by the mother’s ACS person weight. Standard errors are clustered by household, because several eligible women can live in one household, and each outcome is estimated on the mothers for whom it is observed.

With one instrument and one endogenous variable, two-stage least squares is the ratio of the two coefficients:

$$\hat{\beta} = \hat{\delta} / \hat{\pi}. \quad (4)$$

If the instrument is as good as randomly assigned given W_i , affects housing only through the additional child, and moves every mother’s fertility in the same direction, β is the average effect of the additional child for mothers whose fertility responds to the instrument. For the first design these are mothers who, without a same-age pair, would have one coresident child at that event age. For the second they are mothers who have a third child because their first two share a sex. The two groups differ in age, in family size, and in the margin they cross, so the two estimates answer different questions.

Our baseline does not include separate controls for the sex of the first and second child. Angrist and Evans (1998, pp. 460–461) add them because the same-sex indicator is an interaction of the two children’s sexes: with slightly more boys than girls born, it is mildly correlated with each child’s sex, and a child’s sex may affect parents for reasons unrelated to family size. The same authors report that their estimates are virtually unchanged by these controls (pp. 466–467); we do not report that specification here.

We report three measures of uncertainty. Confidence intervals are the estimate plus or minus 1.96 clustered standard errors. The first-stage F statistic is $(\hat{\pi} / \widehat{\text{se}}(\hat{\pi}))^2$ from the clustered covariance; with a single instrument and a single endogenous variable, it coincides with the Kleibergen–Paap statistic (Andrews et al., 2019). Because a large F guards against weak-instrument bias but says nothing about the exclusion restriction, we also invert the Anderson–Rubin test (Anderson and Rubin, 1949), which Andrews et al. (2019) recommend reporting whenever there is a single instrument. For each candidate value β_0 we regress $Y_i - \beta_0 D_i$ on Z_i and W_i and test whether the coefficient on Z_i is zero at the 5 percent level with the same clustered covariance. The candidates form a grid from -3 to 3 in steps of 0.1 for rooms and bedrooms, and from -50 to 50 percentage points in steps of 2 for ownership. The tables report the smallest and largest grid values that are not rejected. With one instrument, the Anderson–Rubin set is the set of β_0 satisfying a quadratic inequality, and it is a bounded interval exactly when the first-stage F exceeds the critical value of 3.84 ; every first-stage F in our tables does. The exact endpoints lie between the last accepted and the first rejected grid value. Where one grid value is accepted, as for rooms in the pooled same-age design, the set lies within one grid step of that value; it is not a point. For bedrooms in the same design no grid value is accepted: the set is narrower than the grid step and lies between two adjacent grid values, in line with the conventional interval of $[0.305, 0.361]$.

A.5 Results

Table 2 reports the estimates pooled over event ages 0 to 5, and Table 3 reports them at event ages 3 and 5. Figure 1 plots all eighteen reduced forms. We report every design, outcome, and window that was estimated.

Table 2: Pooled estimates, event ages 0–5

Outcome	Mothers	Households	Instrument equal to one	Reduced form	First stage (pp)	First-stage F	2SLS	AR accepted grid values
<i>A. Same-age oldest pair, event age 0–5</i>								
Rooms	857,607	855,366	16,496	0.329 (0.017)	66.21 (0.24)	77,396	0.496 (0.026)	{0.5}
Bedrooms	857,607	855,366	16,496	0.221 (0.010)	66.21 (0.24)	77,396	0.333 (0.014)	none
Owner (pp)	857,607	855,366	16,496	3.11 (0.46)	66.21 (0.24)	77,396	4.69 (0.70)	[4, 6]
<i>B. Same-sex oldest pair, event age 0–5</i>								
Rooms	656,985	656,159	329,974	−0.038 (0.005)	3.27 (0.12)	715	−1.149 (0.164)	[−1.4, −0.9]
Bedrooms	656,986	656,160	329,974	−0.034 (0.003)	3.27 (0.12)	715	−1.042 (0.096)	[−1.2, −0.9]
Owner (pp)	656,986	656,160	329,974	−0.25 (0.15)	3.27 (0.12)	715	−7.50 (4.51)	[−16, 0]

Notes: Each row reports the reduced form and first stage of equation (3) and the two-stage least-squares estimate of equation (4), with clustered standard errors in parentheses. The treatment is a second coresident child in panel A and a third in panel B. Rooms and bedrooms are counts, capped at nine and five; ownership and the first stage are in percentage points. “AR accepted grid values” gives the smallest and largest values of the effect not rejected by the Anderson–Rubin test at the 5 percent level on the grid described in the text; braces denote a single accepted value and “none” an accepted set narrower than the grid step. Regressions weighted by the ACS person weight; standard errors clustered by household.

Same-age oldest pair. The first stage is large, 66 percentage points, because a same-age pair guarantees a second child: the coefficient is approximately one minus the probability that a comparable mother without such a pair already has a second coresident child. Mothers with a same-age pair live in 0.33 more rooms (standard error 0.017) and 0.22 more bedrooms, and they are 3.1 percentage points more likely to own. Scaling by the first stage, a second child is associated with half a room, a third of a bedroom, and a 4.7-point higher ownership rate. The first stage falls from 52 points at event age 3 to 35 points at event age 5, as more mothers without a same-age pair have a second child, while the room and bedroom reduced forms change little. The two-stage estimates therefore rise with the event age, to 1.1 rooms and 15 points of ownership at age 5. At event age 3 the ownership reduced form is 1.6 points and its confidence interval includes zero. These magnitudes carry the qualifications above: the proxy mixes twins with closely spaced and blended sibling pairs, and twin births are associated with maternal characteristics that may affect housing directly.

Same-sex oldest pair. The first stage is 3.3 percentage points pooled over event ages 0 to 5 and rises to 4.7 points at event age 3 and 6.0 points at event age 5, consistent with more third births as time passes since the second. With more than 650,000 mothers it is precisely estimated, with an F statistic of 715. The pooled figure is smaller than the 6.2 points that Angrist and Evans (1998,

p. 461) estimate in the 1980 Census, but the samples differ: ours ends five years after the second birth, whereas theirs includes mothers whose second child is up to seventeen, and the two censuses are a generation apart. Mothers whose two oldest children share a sex live in 0.038 fewer rooms (standard error 0.005) and 0.034 fewer bedrooms (0.003), and they are 0.25 percentage points less likely to own, a difference whose confidence interval includes zero. The room and bedroom differences are larger at event ages 3 and 5, about 0.05 and 0.04 respectively, and the ownership differences remain small and imprecise.

The two-stage estimate for rooms is -1.15 , with an Anderson–Rubin set between -1.4 and -0.9 . It is large because the reduced form, though small, is divided by a first stage of 0.033: $-0.0376/0.0327 = -1.15$. Taken literally, a third child would cost the family more than a full room of living space. We do not interpret it this way. The estimate is what one would obtain if same-sex siblings changed housing directly. Under the exclusion restriction the reduced form equals $\pi\beta$, and any direct effect of sibling sex on housing adds to it in full while the first stage stays the same, so the direct effect enters the two-stage estimate multiplied by $1/\pi$, about 31. Jones (2015) shows how such direct effects on mothers whose fertility does not respond to the instrument bias the instrumental-variable estimand. Bedrooms are the natural channel. Two children of the same sex can share a bedroom more easily than a boy and a girl, so a family with same-sex children may need one fewer bedroom at a given size, whether or not it has a third child. Angrist et al. (2010, p. 775) raise economies of scale in room or clothes sharing as a potential violation of the same-sex exclusion restriction, attributing the concern to Rosenzweig and Wolpin (2000). They check it in Israeli subsamples in which sex composition does not predict family size and find no reduced-form relation between sex composition and their outcomes, which concern the children’s schooling and later lives rather than housing (p. 799). The pattern in Table 2, in which the bedroom difference accounts for nearly all of the room difference, is what this channel predicts. It is consistent with the channel but does not establish it, and we do not test the rooms and bedrooms differences against each other, which would require their joint covariance.

The two designs measure different objects, and their estimates should not be averaged or set against each other. Under its assumptions, the same-age design says that a second young child raises the space a family occupies and its likelihood of owning. The same-sex design does not deliver a credible estimate for a third child, because sibling sex appears to act on housing directly, through the composition of the family and not only its size.

Table 3: Estimates at event ages 3 and 5

Outcome	Mothers	Households	Instrument equal to one	Reduced form	First stage (pp)	First-stage F	2SLS	AR accepted grid values
<i>A. Same-age oldest pair, event age 3</i>								
Rooms	147,986	147,922	2,726	0.332 (0.042)	52.46 (0.24)	49,032	0.633 (0.080)	[0.5, 0.7]
Bedrooms	147,986	147,922	2,726	0.212 (0.023)	52.46 (0.24)	49,032	0.405 (0.043)	{0.4}
Owner (pp)	147,986	147,922	2,726	1.63 (1.14)	52.46 (0.24)	49,032	3.11 (2.17)	[0, 6]
<i>A. Same-age oldest pair, event age 5</i>								
Rooms	142,101	142,039	2,498	0.382 (0.041)	34.83 (0.22)	24,972	1.096 (0.119)	[0.9, 1.3]
Bedrooms	142,101	142,039	2,498	0.234 (0.023)	34.83 (0.22)	24,972	0.671 (0.067)	[0.6, 0.8]
Owner (pp)	142,101	142,039	2,498	5.26 (1.15)	34.83 (0.22)	24,972	15.11 (3.30)	[10, 20]
<i>B. Same-sex oldest pair, event age 3</i>								
Rooms	108,847	108,825	54,561	-0.051 (0.012)	4.67 (0.35)	174	-1.089 (0.284)	[-1.6, -0.6]
Bedrooms	108,847	108,825	54,561	-0.040 (0.007)	4.67 (0.35)	174	-0.856 (0.165)	[-1.2, -0.6]
Owner (pp)	108,847	108,825	54,561	-0.53 (0.36)	4.67 (0.35)	174	-11.31 (7.82)	[-26, 2]
<i>B. Same-sex oldest pair, event age 5</i>								
Rooms	92,246	92,229	46,201	-0.053 (0.013)	6.04 (0.41)	214	-0.874 (0.235)	[-1.3, -0.5]
Bedrooms	92,247	92,230	46,201	-0.045 (0.007)	6.04 (0.41)	214	-0.745 (0.137)	[-1.0, -0.5]
Owner (pp)	92,247	92,230	46,201	-0.33 (0.40)	6.04 (0.41)	214	-5.41 (6.59)	[-18, 6]

Notes: See Table 2. Event age is the oldest child's age in panel A and the second-oldest child's age in panel B, in whole years at the interview.

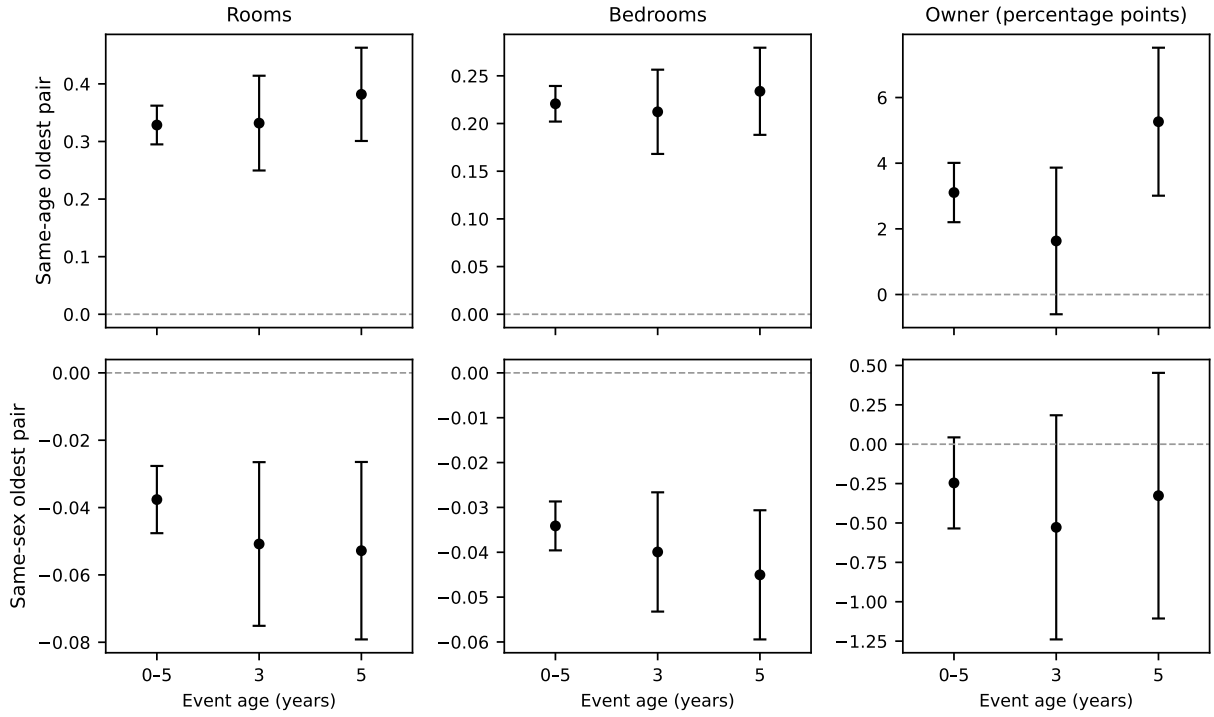


Figure 1: Reduced-form differences in housing by sibling composition

Notes: Each point is the reduced-form coefficient δ of equation (3), with a 95 percent confidence interval from household-clustered standard errors, for event ages 0–5 pooled and for event ages 3 and 5. Top row: mothers whose two oldest linked children are the same age, relative to mothers whose oldest child is the same age but who have no same-age pair. Bottom row: mothers whose two oldest linked children have the same sex, relative to mothers of one boy and one girl. Ownership in percentage points.

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